



FACULTY OF INFORMATION AND COMMUNICATION TECHNOLOGIES
SUMMER 2026
FINAL EXAMINATION SETTING A

SCHOOL	ICT FACULTY
COURSE TITLE/ CODE	CYS 4151 Ethical Hacking
COURSE STATUS	SUMMER 2026
INSTRUCTOR:	Dr. Nkandeu Pascal
DATE:	September 2026
DURATION:	3 Days / 1 Week
VENUE	CISCO LAB

SCENARIO

SecureTech Solutions Ltd. – Enterprise Security Assessment

SecureTech Solutions Ltd. is a growing software development and digital services company with approximately 120 employees. The company provides web applications, cloud-based services, and remote access for its staff. Recently, management observed several security incidents, including:

- Multiple unauthorized login attempts against internal systems.
- Employees using weak or reused passwords.
- Suspicious activities detected on the company web server.
- Sensitive customer information potentially exposed.
- Unpatched systems and outdated software.
- Lack of centralized monitoring and security documentation.

To strengthen its cybersecurity posture, the company has hired your cybersecurity consulting team to perform a comprehensive Ethical Hacking and Security Assessment.

You have been granted written authorization to conduct all testing only within the provided laboratory environment.

Your assignment is to simulate the activities of both a Red Team (offensive security) and a Blue Team (defensive security) by identifying vulnerabilities, demonstrating controlled exploitation, analyzing evidence of attacks, and proposing effective remediation strategies.

The objective is not only to compromise systems but also to demonstrate professional cybersecurity assessment, risk analysis, documentation, and defensive recommendations.

Laboratory Infrastructure

Your laboratory should contain at least the following systems:

System	Purpose
Kali Linux	Attacker machine
Windows 10/11 Workstation	Employee computer
Windows Server or Ubuntu Server	File/Application Server
DVWA or OWASP Juice Shop	Vulnerable Web Application
Metasploitable2 or Metasploitable3	Vulnerable target
pfSense (optional)	Firewall/Router
Wireshark	Traffic analysis
OpenVAS/Nessus/Greenbone	Vulnerability assessment
VirtualBox or VMware	Virtual laboratory

Students may enhance the environment with additional systems where appropriate.

GENERAL REQUIREMENTS

Students must conduct all activities ethically and only inside the authorized laboratory environment.

Every submission must include	Video Requirements	Deliverables
- Professional Technical Report (PDF) - Network Architecture Diagram - Vulnerability Assessment Report - Risk Assessment Matrix - Security Hardening Recommendations - Screenshots demonstrating every major task One high-quality video for each task	Every video must clearly show: - Student name - Student ID - Hostname or computer name - Username - Date and time - Commands executed - Results obtained Videos must be clear, continuous, and narrated where appropriate.	Students shall submit: - Professional Penetration Testing Report (PDF) - Network Architecture Diagram - Asset Inventory - Vulnerability Assessment Report

Final presentation slides (optional if requested)		5. Risk Register 6. Security Hardening Plan 7. Screenshots supporting every task 8. One video per task 9. Source files (where applicable)
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Failure to verify the identity of the candidate or evidence of manipulation may result in **cancellation of the examination**.

PROJECT TASKS (TOTAL = 70 MARKS)

No.	Task	Description	Marks
1	Laboratory Design & Documentation	Build the virtual lab, configure networking, assign IP addresses, produce a professional network topology diagram, and justify the architecture.	5
2	Reconnaissance & Asset Discovery	Perform host discovery, identify operating systems, IP addresses, network devices, and prepare an asset inventory.	4
3	Service Enumeration	Identify running services, versions, open ports, protocols, web technologies, and potential entry points using appropriate tools.	4
4	Threat Modeling & Risk Identification	Identify critical assets, possible attackers, attack vectors, security threats, and assess the associated business risks.	4
5	Vulnerability Assessment	Perform a vulnerability assessment using automated and manual techniques. Identify at least five significant vulnerabilities, classify their severity (e.g., CVSS), and provide supporting evidence.	6
6	Controlled Exploitation (Initial Access)	Demonstrate the exploitation of at least one confirmed vulnerability to obtain controlled access. Explain the attack path and maintain professional documentation.	6
7	Web Application Security Assessment	Assess the vulnerable web application and successfully demonstrate at least two OWASP Top 10 vulnerabilities (e.g., SQL Injection, XSS, Broken Authentication, File Upload, Command Injection).	6
8	Credential Security Assessment	Evaluate password security by auditing password policies and demonstrating password attacks within the lab (e.g., password guessing, brute force, or hash cracking). Provide recommendations for stronger authentication.	4

9	Privilege Escalation & Persistence Analysis	Demonstrate privilege escalation where applicable, explain why it was possible, and discuss persistence mechanisms that attackers might use without deploying malicious persistence outside the lab.	5
10	Post-Exploitation & Lateral Movement Analysis	Demonstrate how an attacker could enumerate resources and assess opportunities for lateral movement from a compromised host. Document findings and associated risks.	5
11	Blue Team Investigation (Traffic & Logs)	Capture network traffic using Wireshark and analyze relevant operating system or application logs to identify indicators of compromise and explain how the attacks can be detected.	5
12	Security Hardening & Incident Response	Develop a remediation plan including system hardening, firewall rules, secure configurations, patch management, password policies, monitoring recommendations, and a brief incident response workflow.	5
13	Professional Penetration Testing Report	Produce a complete professional report including Executive Summary, Scope, Methodology, Findings, Risk Ratings, Evidence, Business Impact, Recommendations, and Conclusion.	11
	TOTAL		70

EXPECTED METHODOLOGY

Students are expected to follow a professional penetration testing methodology similar to PTES, NIST SP 800-115, or the OWASP Testing Guide.

A logical workflow should be demonstrated: Planning, Reconnaissance, Enumeration, Threat Modeling, Vulnerability Assessment, Controlled Exploitation, Privilege Escalation, Post-Exploitation Analysis, Traffic & Log Analysis, Risk Assessment, Security Hardening, Professional Reporting.

ALL THE BEST

